

Data 780 Project Proposal

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Objective: To build a model that can accurately predict World Cup results using historical data. We propose to project each team's goals scored and goals allowed per game played in this summer's tournament.

Methods: Broadly, we will use a combination of historical results in international competition, alongside data about the specific players on each team's roster, in order to derive our projections. The question of which model types and data features are most predictive will be a major focus of our research. We will retrospectively analyze results from past World Cups to evaluate different model choices.

Datasets:

To have statistically significant results, we plan to utilize the past 8 years of international men's match results and player statistics as training data for our prediction model. For this project, this will be utilizing the following datasets:

- For international results, this will be utilizing the ['International Football Results from 1872 to 2026' Kaggle dataset](#) by Mart Jurisoo. This dataset captures detailed data on match results and statistics. This dataset is also CC0-1.0 licensed which makes this public domain and gives credence to further reproducibility.
- For player statistics, our best source appears to be [Football Reference](#). Football Reference discourages data scraping from its site, so a question we will need to solve is how best to source and aggregate this data.
- As our project continues, we will be matching our project results to the current FIFA world cup to see how our model predicts against the current tournament matchups.

Evaluation Criteria: From our initial research, the most detailed publicly available World Cup forecasting model appears to be [Nate Silver's 'Pele' system](#), which released its [World Cup predictions](#) on June 4th, 2026. We hope to build a model with predictions that perform better than Silver's system in this tournament.

Challenges & Drawbacks: Our initial research suggests that compiling player-level statistics for every roster, and wrangling that data into a usable form, will be the most challenging aspect of our proposed approach. Many systems designed to track international soccer **only** use historical match data because modeling active rosters adds so many complications. However, we believe this player-level data has the possibility to make a model more predictive. We are investigating possible sources for aggregating players' club statistics; this remains an open question for us.

Framework/Language: We plan to use Python; primarily scikit-learn and TensorFlow, most likely. As the project continues, we plan to explore different supervised learning methodologies such as OLS regression models, decision trees and random forest to determine which methods will provide the best predictions.